Pre-registered protocol: a global sensitivity audit of the scenario-generation pipeline of Li et al. (2025), *Nature Climate Change*

Modelling of the Modelling Process (MoMP) series: optimal-transport GSA case

Version 0.5 (release candidate), [TBD: date of deposit], DOI: [TBD: reserved on Zenodo before upload]

Release candidate. Values agreed in the co-authors' review of 23 October 2026, and the items of v0.3 and v0.4 approved in reply, are in black. Text in blue awaits approval. New in v0.5 are Phase 0E and D15 (the 901-set contains no IMAGE or WITCH scenario, because none of their 257 vetted scenarios reports geothermal electricity at World level), the model groups of X5 with their AR6 versions (Appendix A), the fixed folds, per-category gates and out-of-fold reference mixes of X5, and the specifics of the comparison with the 2023 case, whose results file is archived with this deposit (artefact C1). Items in [TBD:] are filled at deposit. This box, all colour and all placeholders disappear in the frozen version.

Abstract

We pre-register a global sensitivity analysis (GSA) of the scenario-generation pipeline of Li, Zhu, McJeon, Byers, Zhou and Ou (2025), "Using deep learning to generate key variables in global mitigation scenarios", *Nat. Clim. Change* 15:760–768 (`doi:10.1038/s41558-025-02352-8`). The object of analysis is not the climate or the AR6 ensemble but the pipeline itself. The analyst's methodological choices are the input factors, the generated multivariate ensemble and the paper's validation claims are the outputs, and sensitivity is measured by optimal-transport (Wasserstein) displacement of the synthetic ensemble, estimated from given data. Before any GSA run, this protocol freezes the operational rules for the paper's claims (§4), the factor space (§5), the experimental design and compute budget (§6), the estimators and inference procedure (§7), and the decision criteria (§8). The completed replication record (Phase 0, §3) is reported for transparency. At the time of this deposit, no claim rule (X1a–X5) has been evaluated at any factor setting, including the paper's own values, no retraining has been performed, and no noise-floor replicate has been run. The only pipeline stage executed is the deterministic regeneration check of Phase 0B, which evaluates no claim. The analysis code will be deposited as a new version of this record before the first run (§10).

1 Purpose and scope

Continuing the MoMP programme, this study asks whether the properties Li et al. attribute to their synthetic-scenario framework are properties of the framework or of the specific, partly undocumented methodological choices through which it was instantiated. We audit the epistemic apparatus, not the climate–economy content: factors are analyst choices, outputs are (i) the synthetic ensemble as a multivariate distribution and (ii) pass/fail evaluations of the paper's claims under fixed operational rules.

2 Audited artefacts

All artefacts are pinned by version and SHA-256 in `DATA.md` of the project repository, archived with this deposit, the Deep-IAM deposits file by file in the unpacked form read by the scripts

and, as archives, by Zenodo's MD5. The principal items follow.

A1 Deep-IAM code, v1.0.0, `doi:10.5281/zenodo.15280653`, the version cited by the paper and the audited object.

A2 Deep-IAM v1.0.1 (2026-04-25), admitted as a documented supplement only for its two added files: `Secondary-data-generate.ipynb` (the generator of the deposited synthetic data, absent from v1.0.0) and `Configuration of Environments.txt` (environment pins: Python 3.8.20, TensorFlow 2.13.0, tsgm 0.0.7, TensorFlow Probability 0.21.0, scikit-learn 1.3.2, XGBoost 2.1.1). Full diff archived.

A3 Synthetic-scenario deposit, `doi:10.5281/zenodo.15240553` (Gen and Real CSVs, validated in Phase 0).

B1–B2 AR6 Scenarios Database v1.1 (metadata and World timeseries), `doi:10.5281/zenodo.7197970`.

C1 Results of the earlier case of the series, the analysis of Li P.-H. et al. (2023) with Jansen's estimators, with the script that generated them and a README, archived under `data/external/gsa2023/` for the comparison of §7.

3 Phase 0: completed replication record

Completed before this deposit. Machine-readable reports are archived under `results/`.

1. **Metadata fidelity.** The AR6 v1.1 vetted-with-climate population reproduces the paper's Extended Data Table 1 exactly: $n = 1202$, with per-category counts matching digit for digit (`phase0_metadata_check.json`).
2. **First undeclared filter reconstructed (D2).** The training population of SI S3.1 (680/308/172, $n = 1160$) is reproduced exactly by complete 2020–2100 coverage on the five feature variables (`PrimaryEnergy|Coal`, `|Oil` and `|Gas`, `FinalEnergy|Liquids`, `CarbonSequestration|CCS`), with Kyoto Gases exempt. This rule appears nowhere in the paper or SI.
3. **Second undeclared filter reconstructed (D9).** The population underlying the paper's figures and label-transfer benchmark (Fig. 4a, ED Fig. 2, Table 1: 554/237/110, $n = 901$) is reproduced exactly by complete 2020–2100 coverage on the five feature variables and all ten electricity variables (the total and nine fuels), Kyoto Gases exempt, raw (pre-imputation) CS availability required. Across the two silent drops the smallest category (C78) loses 43% of its vetted scenarios (193→110).
4. **Deposit validated.** After harmonizing a Mt-versus-Gt unit inconsistency (deposit in Mt, the paper's Source Data Fig. 3 in Gt), all 1,620 quantile comparisons (6 variables × 6 series × 9 years × 5 quantiles) pass at 2% relative tolerance. Gen files hold 10,000 scenarios per category.
5. **Version provenance (D7).** v1.0.0 cannot reproduce the deposited synthetic data or Fig. 5. The generating notebook appeared in v1.0.1 ten months after publication, without changelog.
6. **Non-reproducibility of training (D1).** The authors state that TSGM training is not reproducible even under fixed seeds and distribute trained weights, which motivates the noise-floor design of §6.
7. **Phase 0B, executability and determinism of the generation stage.** Evaluating no claim rule, we executed the pipeline's one nominally deterministic link. The authors' shipped decoder weights (`vae_conv5`, 9 steps × 16 features, latent dimension 10) were loaded, the `TSFeatureWiseScaler` was refit on the deposited Real CSVs (which the D9 reconstruction identifies as the training tensors), the latent noise was drawn under the documented global seed 3, decoded, and compared with the deposited Gen CSVs (environment: Python 3.10.21, TensorFlow 2.13.0, tsgm 0.0.7, macOS/arm64, whereas the authors used Windows x86-64 and Python 3.8; report `phase0b_regeneration.json`). Executability is confirmed: the weights load, the architectures match and generation runs end to end from

the archived artefacts alone. Sample-level reproduction fails under all three latent-stream readings of the documentation (TensorFlow global seed, a second-execution probe and the SI's NumPy reading), with median elementwise relative deviation 0.33–0.45, i.e. entirely different draws. The distributions agree in the bulk, with median quantile deviation 1.8% of the reference spread (Q90–Q10) and 95th percentile 10.3%, yet about 4.4 times the Monte Carlo floor obtained from two independent 10,000-draw ensembles through the same decoder, the excess being concentrated in distribution tails and tight-spread cells (e.g. carbon sequestration in 2020). A scaler-bounds diagnostic flags no feature, consistent with the Real CSVs being the scaler-fit data. The residual is most plausibly a checkpoint mismatch (the shipped weights differing from those that generated the deposit), consistent with the repository README's account of multiple training sessions with "comparable" results. The generation stage is therefore executable and distributionally faithful, but the deposited outputs cannot be tied to the documented seed or, apparently, to the shipped weights. The determinism gap the authors acknowledge for training (D1) extends to generation (D11). This check answers no question posed in §4–§8.

8. **Phase 0C, static adjudication from code and real data.** Without executing any model, the v1.0.0 notebooks were scanned (`phase0c_code_scan.txt`) and the AR6 reference value of the electricity sum-check was recomputed from the deposited Real CSVs alone (`phase0c_sumcheck_reference.json`). No synthetic file was read. The SI's training settings are those of the five-feature "Top6" notebooks and Extended Data Table 3's those of the sixteen-variable "Secondary" notebooks, up to two slips (batch size 25, not 30, for the C78 VAE and decay steps 1,000, not 200, for the RCGAN), which resolves D4 in favour of the code. Every generative notebook runs its own label-transfer test, and the test-set sizes of Extended Data Table 2 (554/237/110) occur only in the Secondary notebooks, so the published comparison of the three generators was like-for-like (D12). That test uses a random forest tuned by three-fold grid search, 1,000 synthetic scenarios per category and, as inputs, the trapezoid-cumulated 2020–2100 values of fifteen variables, Kyoto Gases excluded. The deposit also holds unreported variants of all three generators with seven to nine variables, added in the order of the eight-category ranking of SI Fig. S3b and without saved outputs (D14). No code for the electricity sum-check exists in either version of the deposit. Recomputed on the 901-set, the AR6 average error is 1.43% when each scenario's error is averaged over the nine decades, and none of six aggregations fixed before the computation rounds to the paper's 1.2%. The distribution is highly skewed: half of all scenario-decades have errors below 0.011%, so the average is driven by a minority of scenarios (D13).
9. **Phase 0D, denominators of the normalizations.** From the deposited Real CSVs alone, the real 10–90 bands that normalize X1a and the standard deviations that standardize the ensemble marginals were computed per variable, category and decade (`phase0d_denominator_guard.json`). No synthetic file was read. No band or standard deviation is zero, but some are small enough to make the ratios ill-conditioned. Carbon sequestration has bands below a tenth of its median band over the decades in five cells (2020 in all three categories, 2030 in C56 and C78), where 20 to 55% of the scenarios report zero. Thirteen of the 432 variable–category–year marginals of the ensemble distance, all in 2020, have standard deviations below a tenth of their variable's median, because the scenarios are harmonized to the base year or, for carbon sequestration, mostly report zero, and would magnify synthetic spread up to 262 times relative to the variable's typical decade. The floors of §4 and §7 follow from this check.
10. **Phase 0E, model groups.** From identifiers (model, scenario and category in the AR6 metadata and the two derived scenario lists), and from whether each of the ten electricity variables is reported in each decade of the World CSV, the AR6 model names were mapped to the frameworks of SI Table S3 and counted in the 1,202 vetted scenarios, the 1,160 of D2 and the 901-set (`phase0e_model_groups.json`). No scenario value enters the report, and no synthetic file was read. Of the 259 scenarios that the D9 rule removes from the 1,160,

258 do not report geothermal electricity: all 142 IMAGE and all 115 WITCH scenarios and the single MERGE-ETL one. The remaining one, from MESSAGE, does not report total electricity. The 901-set on which the generators are trained therefore contains seven of the nine model groups of SI S2.3, together with GEM-E3, a model outside them. IMAGE and WITCH also account for 62 of the 83 C78 scenarios lost between the 1,202 and the 901-set. The groups and counts are in Appendix A (D15). The same script fixes the five folds of the X5 classifier from identifiers alone (`data/derived/x5_folds_901.csv`, §4).

The full inconsistency catalogue (D1–D15) is archived as `DISCREPANCIES.md`.

4 Claims and operational rules

Each claim of the target paper is translated into a rule verifiable on the pipeline's output. Populations are declared explicitly, given D2 and D9. Claims are evaluated on the three aggregated categories (C1234, C56, C78) and on the evaluation set E of §7. When factor G2 uses finer categories, each finer category is trained with the settings of the aggregate it belongs to, receives a share of the G7 sample size proportional to its real count in the 901-set, and its synthetic ensemble is pooled into that aggregate. The three k-means clusters are mapped one-to-one to the three aggregates by the assignment that maximizes the number of real scenarios they share. No rule will be altered after deposit except under the deviations policy (§9).

X1a (distributional match).
"[The VAE] closely matches the original AR6 scenarios in terms of mean and the spread, particularly for C1234." Rule: for each of the six key variables and each category, the median, over the nine decades and the five quantiles (Q10, Q25, Q50, Q75, Q90), of the absolute difference between synthetic and real quantiles divided by the real 10–90 percentile range of that variable in that category and decade (the band drawn in the paper's Fig. 3, never pooled across categories) is at most $\tau_1 = 0.15$, the real population being the 901-set. A band smaller than a tenth of the median of that variable's bands over the nine decades, in the same category, is replaced by that tenth. Phase 0D found this floor binding in five of the 162 cells, all carbon sequestration (2020 in all three categories, 2030 in C56 and C78). "Particularly C1234" holds if the C1234 deviation, pooled over variables, decades and quantiles (the median of its 270 elements), is the smallest of the three categories. X1a holds if all 18 variable–category medians are at most τ_1, and "particularly C1234" is recorded as a separate indicator, X1a$'$, with its own pass fraction and indices. This normalization replaces the deviation relative to the quantile itself, which is ill-conditioned where quantiles approach or cross zero (coal in C1234 late in the century, Kyoto gases in C1234 around net zero, carbon sequestration in C78). In the Phase 0B check, a synthetic-versus-synthetic comparison, relative deviations reached about 1,260% (electricity from coal, C1234, 2070, Q10) in a category where no spread-normalized deviation exceeded 0.34. The floor keeps the ill-conditioning from moving to the band. The pass fraction is also reported with floors of 5% and 20%, and the maximum deviation over the 45 elements of each variable and category is reported next to the median, since the median is insensitive to misfit in the tails. Neither enters a decision.

X1b (narrower distributions).
"Synthetic scenarios have narrower distributions for all climate categories." Rule: $\mathrm{IQR}_{\mathrm{syn}} < \mathrm{IQR}_{\mathrm{real}}$ for all six key variables and all three categories in 2050 and 2100.

X2 (correlation replication).
"The synthetic scenarios also retained the correlations between key variables … matching closely those observed in the original AR6 scenarios." Rule, for C1234, the category shown in the paper's Fig. 4: (i) the maximum absolute elementwise difference between the synthetic and real Pearson correlation matrices of the six key variables, taken over the 15 pairs and

the three years 2030, 2050 and 2100 (45 elements), is at most $\tau_2 = 0.15$, and (ii) the paper's reported temporal ordering (coal–Kyoto correlation stronger in 2030 than in 2100) holds in the synthetic data, i.e. the signed Pearson correlation of coal and Kyoto Gases in 2030 exceeds that in 2100 (0.82 and 0.21 in the real 901-set). The sampling noise of a correlation is at most about $1/\sqrt{n}$, i.e. about 0.04 at $n = 554$, hence about 0.13 on the maximum of 45 elements, and larger in practice because scenarios from the same IAM are not independent. τ_2 is a heuristic ceiling, since the real 901-set is the training data and the generator fits that sample.

X3 (internal consistency).
"Among the 30,000 synthetic scenarios, 258 exhibited lower sum-check errors than the 1.2% average error in the AR6 Scenarios Database." Rule: the fraction of synthetic scenarios with electricity sum-check error below 1.2% is at least 258/30,000 = 0.86%. The sum-check error of a scenario is $|\mathrm{Gen}_{\mathrm{total}} - \sum_i \mathrm{Gen}_i|/\mathrm{Gen}_{\mathrm{total}}$ (the paper's Eq. 3), aggregated over the nine decades by the mean. Phase 0C found no code for the sum-check and no aggregation that reproduces the 1.2% reference on the 901-set (D13), so the threshold stays at the value the paper states. Stating the benchmark as a fraction prevents the sample-size factor (G7) from moving the rule mechanically.

X4 (VAE superiority).
"VAE achieved higher category label transferring accuracy ... compared to RCGAN and CGAN" (Experiment I, synthetic→real, test population the 901-set). Rule: accuracy(VAE) > accuracy(RCGAN) and accuracy(VAE) > accuracy(CGAN) in the same evaluation. As in the authors' code, the classifier's inputs are the trapezoid-cumulated 2020–2100 values of the fifteen variables of E other than Kyoto Gases, and the synthetic training set has 1,000 scenarios per category. The three generators are trained together, on the same population and the same variables, so the comparison is like-for-like by construction, as Phase 0C shows the published one was (D12). G7 does not act on X4.

X5 (bias inheritance, secondary).
The paper positions synthetic generation as mitigating "model-determined biases" and "non-random submission". Rule: a source-model classifier (model versions merged into frameworks as in SI S2.3, which follows Dekker et al. (2023), *Nature* 624:309–316, on the eight groups present in the 901-set; Appendix A, D15) is trained on the real 901-set and applied to the synthetic scenarios. X5 (inheritance) holds if, in each category that passes its gate, the total-variation distance between the predicted model mix of the synthetic ensemble and the real reference mix is smaller than the distance to the uniform reference mix. A baseline model-identifiability check on real data (held-out accuracy above the majority-class rate) gates the test. The classifier is the random forest of G8's paper level, trained in the five folds of the real 901-set with the trapezoid-cumulated 2020–2100 values of the sixteen variables of E as inputs. The gate is passed if the lower bound of the 95% Clopper–Pearson interval of its five-fold stratified cross-validated accuracy exceeds the majority-class rate, 296/901 = 0.329 (REMIND). If the gate fails, X5 is reported as not testable and has no indices. G7 sets the size of the synthetic ensemble classified, and G8 does not act on X5. The folds are fixed in this deposit from identifiers alone (Phase 0E, `data/derived/x5_folds_901.csv`): the 901 scenarios are ordered by category, group and the SHA-256 of `model|scenario` and dealt to folds 1 to 5 in turn, so that each category, and each group within it, is spread evenly. Each fold is predicted by a classifier fitted on the other four, with the grid search of G8's paper level inside them and rows in the order of the SHA-256 key, which also fixes the grid search's inner split. No classifier is refitted on the whole 901-set, and accuracy counts correct out-of-fold predictions. The fold classifiers, gates and references are computed once with the deposited code, after the code deposit and before the first run, and archived. The same gate is applied within each category,

against that category's majority rate (C1234, REMIND, 0.334; C56, MESSAGE, 0.325; C78, REMIND, 0.527). A category that fails its gate is reported as not testable, and X5 is not testable if the pooled gate fails or no category passes. A mix is a vector of shares of hard labels. A classifier shrinks its predictions towards the larger groups, and its behaviour differs between categories, so both references are formed within the category and through the same classifiers as the synthetic scenarios. The real reference of a category is the mix of the out-of-fold predictions for its real scenarios. Its uniform reference is the mix predicted for a uniform mix of the eight groups, i.e. the mean of the rows of that category's out-of-fold confusion matrix, rows being true groups, each normalized to sum to one. The same reference computed from the confusion matrix of the whole 901-set is reported as a variant. The synthetic mix of a category is the mix of the labels that the five fold classifiers assign to its synthetic scenarios, averaged over the five. Under every factor level the references are those of the 901-set, the real comparator of every claim. Two limitations are stated in advance. Each fold classifier has seen four fifths of the real scenarios the generator learns from, so it may label synthetic scenarios with less shrinkage than held-out real ones. Scenarios of the same model and study can fall in different folds, so out-of-fold accuracy and the gates are optimistic.

5 Factor space

Factors, paper values and perturbation ranges are frozen in `config/paper_values.yaml`, archived with this deposit and summarized in Table 1. Expensive factors require retraining, cheap factors vary per trained model. Where the configuration file and this protocol disagree, the protocol governs.

Table 1: Factor space: paper values with provenance, ranges and cost.

Factor	Paper value (provenance)	Range or levels	Declared	Cost
G1 population	901-rule (D9, undeclared; see D12 and D15)	{901; 910, CS imputed; 828, Kyoto required; 1,160, D2 rule; 1,202, all vetted}, missing values filled by G4	no	exp.
G2 aggregation	C1234/C56/C78 (Methods)	{3-way; 4-way; 8-way; k-means labels}, pooled for evaluation (§4)	yes	exp.
G3 variable selection	5 key features ranked by RF (SI S2)	pool completeness threshold 0.6–0.9; number of key features 4–9, generated set united with E	yes	exp.
G4 preprocessing	mean imputation by year × category; (0,1) scaling	imputation {mean yr×cat.; mean yr; median yr×cat.; none}; scaling {(0,1); standardized}	yes	exp.
G5 architecture	VAE latent dimension 10; training settings of the Secondary notebooks (ED Table 3 up to two slips, D4)	latent dimension 4–20; epochs × 0.25–2, all three generators	partly	exp.
G6 seed	seed 3; training non-reproducible (D1)	integer seed, uniform; an ordinary factor (§7)	yes	exp.
G7 sample size	10,000 per category, for X1–X3 and X5 (X4 uses 1,000, as in the code)	1,000–30,000 per category	yes	cheap
G8 transfer classifier	RF tuned by three-fold grid search, random state 42 (code)	{RF as in code; RF defaults, no grid search; logistic; XGBoost}	partly	cheap

Factor G9 (validation metrics) of the circulated version is withdrawn: its range was never specified, and varying the thresholds of the claim rules would turn the pre-registered rules into moving targets. The thresholds are frozen in §4, and their influence is reported descriptively (§7). The

D4 documentation conflict (CGAN and RCGAN epochs, SI versus ED Table 3) is resolved by Phase 0C: the paper's configuration uses the Secondary notebooks' settings, so no nested factor is needed. The design also carries four dummy columns that are never passed to the pipeline, D1–D3 drawn with G1–G6 and D4 with G7 and G8. They are the null reference of §8. Continuous factors are drawn uniformly on their ranges, except the epochs multiplier and the sample size, which act multiplicatively and are log-uniform. Integer factors are uniform on their integers, and categorical levels are equiprobable. The column Declared records whether the paper or its SI states the paper's value: the population filters are undeclared (D2, D9), the training settings are declared up to the slips of D4, and the classifier is declared while its tuning appears only in the code. G1 and G4 interact by construction: under G4's level none, scenarios with missing values are dropped, so every G1 level shrinks to its complete cases on the generated set, which are the 828-set when the generated set is E.

6 Design and compute budget

1. **Noise floor.** $R_0 = 20$ seed-only replications at paper values for all factors. The floor δ_0 is the 95th percentile of pairwise distances between replicate ensembles, per category, in the marginal distance of §7 with its floors. The replicates also give the pass rate of every claim at the paper's own configuration, i.e. the statistical replication of the published results.
2. **Outer design.** Scrambled Sobol' quasi-random sample of size $N_{\text{outer}} = 512$ over the expensive factors G1–G6 and the dummy columns D1–D3, one training per point, environment pinned to A2's configuration.
3. **Inner design.** Per trained model, $N_{\text{inner}} = 8$ points of an independently scrambled Sobol' sequence over the cheap factors G7 and G8 and the dummy column D4. Each trained generator produces 30,000 scenarios per category once, subsampled to the G7 value of each inner point. A training is one outer point and an evaluation one inner point of one training, 4,096 in all.
4. **Stopping rule.** The budget is $N_{\text{outer}} \times$(training cost). At $N_{\text{outer}}/2 = 256$ trainings the 95% bootstrap intervals of the primary index (§8) of every expensive coordinate and of the matched dummies are computed. If all have half-widths below 0.05, the design stops there, otherwise it runs to N_{outer}. It may not be extended beyond N_{outer} without a documented deviation.
5. **Outputs stored per training:** per category, the standardized quantile vectors on E, the 200 projected quantile vectors and a 2,000-point subsample of the 10,000-scenario ensemble, all claim evaluations X1a–X5 for each inner point, wall-clock time, and the full factor vector.

7 Estimators

Ensemble-level sensitivity. Each training yields, per aggregated category, an ensemble of 10,000 synthetic scenarios (G7's paper value) represented on the evaluation set E: the six key variables and the ten electricity variables over the nine decades 2020–2100. Each variable–year marginal is standardized by its standard deviation in the corresponding category of the real 901-set, floored, as in X1a, at a tenth of the median of that variable's standard deviations over the nine decades in the same category. The floor binds for 13 of the 432 variable–category–year marginals, all in 2020, where the scenarios are harmonized to the base year or, for carbon sequestration, mostly report zero (Phase 0D). Without it, synthetic spread in those marginals would be magnified up to 262 times relative to the variable's typical decade, and 2020 could dominate the distance. Distances and indices are also reported with floors of 5% and 20%. Quantiles are computed by linear interpolation between order statistics, and standard deviations with the divisor $n - 1$. The distance between ensembles is the one-dimensional Wasserstein distance on each variable–year marginal, aggregated by the mean of squared Wasserstein-2 distances and square-rooted.

It is computed on the quantiles at the 99 levels $(k-0.5)/99$, $k=1,\dots,99$, so that it is, up to a constant, the Euclidean distance between the vectors of standardized quantiles, and the outer half per cent of each tail is not represented. It is complemented by the sliced Wasserstein distance on the joint, with $L=200$ projection directions fixed once, shared across all trainings and replicates and archived with the harness, so that projection noise is not confounded with training noise. Both distances embed the output Y of each training as a point of a Euclidean space. The sensitivity index of factor X_j is the given-data optimal-transport index of the Borgonovo–Chiani–Plischke class with quadratic cost, applied to one-dimensional projections of Y. For $L'=200$ unit directions u_l of the embedding space, drawn once and archived like the slicing directions,

$$\iota_j \;=\; \frac{\sum_{l=1}^{L'} \mathbb{E}\big[W_2^2\big(P_{u_l^\top Y},\, P_{u_l^\top Y\mid X_j}\big)\big]}{\sum_{l=1}^{L'} \mathbb{E}\big(u_l^\top Y - u_l^\top Y'\big)^2},$$

each W_2 being the exact one-dimensional distance between the empirical distributions of all trainings and of the trainings in one cell. The index lies in $[0,1]$, vanishes under independence and equals one when Y is a function of X_j. It is first-order in that it conditions on X_j alone, and it responds to changes in the spread and shape of the output, not only in its mean. The projection is what makes the index estimable at this budget. The embedding has tens of thousands of coordinates, and in simulations with 20 to 2,000 dimensions the given-data estimator of the unprojected index takes null values of 0.45 to 0.73 for eight cells at $N=512$, approaching $1-1/m$ for m cells, against about 0.02 for the projected index. Because both embeddings are Euclidean, the variance-based indices aggregated over the coordinates of Y (Gamboa et al. 2014) are also estimated, for the secondary criteria of §8: the first-order index $\operatorname{tr}\operatorname{Cov}(\mathbb{E}[Y \mid X_j])/\operatorname{tr}\operatorname{Cov}(Y)$ with the same partitions, and the total-order index $\mathbb{E}\operatorname{tr}\operatorname{Cov}(Y \mid X_{\sim j})/\operatorname{tr}\operatorname{Cov}(Y)$ with the nearest-neighbour estimator and the caveats of the claim-level paragraph. Conditional laws are estimated with $M=8$ bins at $N=512$ (about $N^{1/3}$, 64 trainings per bin), a heuristic choice, with results also reported for $M\in\{5,8,12\}$. Discrete factors use their levels as cells. Concretely, the cells number 2 (G4 scaling), 4 (G2, G4 imputation, G8), 5 (G1), 6 (G3 number of key features) and M equal-count bins (G3 pool completeness, the G5 epochs multiplier, G6 and G7). The latent dimension's 17 integer values form M cells of consecutive values with sizes as equal as possible, the larger ones last, so for $M=8$ the cells are 4–5, 6–7, …, 16–17 and 18–20. With equiprobable levels every cell holds at least a twelfth of the trainings for $M\le 12$, so no cell is merged. Indices are computed per aggregated category and for the three categories jointly, the joint embedding concatenating the three categories' vectors. As a robustness check, the debiased Sinkhorn divergence is computed with $\varepsilon=0.05$ times the median pairwise cost (squared Euclidean distance in the standardized space), computed on subsamples of 2,000 points per ensemble. ε is computed once, on the deposited synthetic ensembles (A3), and held fixed for all comparisons. The index is then obtained with this divergence as the ground cost between trainings. A divergence admits no projection, so this variant keeps the large null value of the unprojected index and is read against its matched dummies only.

Null reference. The dummy columns D1–D3 (outer design) and D4 (inner design) are drawn with the factors and never passed to the pipeline, so every index computed on them is a pure null. Each dummy is cut into cells with the same proportions as each factor it serves, and the reference for a factor is the mean index of the dummies of its design level, cut like that factor. A single reference would not do, because the null value of a binned estimator grows with the number of cells: about $(m-1)/N$ for the variance-based indices, i.e. 0.002 for two cells and 0.014 for eight at $N=512$, and of the same order for the projected OT index (0.004 and 0.02 in simulation). The training seed is an ordinary factor. Each seed is used once, so any effect of the seed, if training is partly seed-controlled, is an erratic function of its value that no binned or nearest-neighbour estimator can resolve. Its index therefore behaves as a null either way and cannot show whether seeds control training.

Claim-level sensitivity. For each rule X1a–X5, the fraction of evaluations in which the claim holds, over the full design and, separately, over the R_0 replicates at the paper's values. Given-data first-order indices S_j of the binary pass indicator are estimated with the same cells, as the between-cell variance of the pass rate over its total variance, for all eleven coordinates. Total-order indices T_j are estimated from the same evaluations with the nearest-neighbour estimator of the residual variance $\mathbb{E}[\mathrm{Var}(Y \mid X_{\sim j})]$ (Devroye et al. 2018; Broto et al. 2020). Neighbours are found with the Gower distance on the coordinates of $X_{\sim j}$, with equal weights, numeric coordinates scaled by their range, and categorical coordinates and the seed counted as mismatches. Dummy columns never enter a search. For an expensive coordinate a neighbour is never taken from the same training, nor in a bootstrap resample from a copy of it. The nine expensive coordinates vary only across the 512 trainings, so an evaluation's nearest neighbour is typically distant in them, which biases T_j upward. Because training is not reproducible (D1), every T_j also contains the share of variance due to training noise. The reference total T_D is the same estimator conditioned on all factor coordinates, the share of variance that no factor explains, i.e. training noise plus the estimator's bias. Since dummies never enter a search, it is common to D1–D3. For G7, G8 and D4 neighbours are taken within the same training, so their totals are conditional on the trained generator, and the reference for G7 and G8 is T_{D4}, computed in the same way. Total-order indices are exploratory and enter only the secondary criteria of §8. The residual $1 - \sum_j S_j$ contains interactions and training noise and is read as set out there.

Threshold sensitivity. The pass fraction of each claim is also reported as a function of its threshold (τ_1 and τ_2 from 0.05 to 0.30, sum-check threshold from 0.01% to 2.0% on a logarithmic scale, which spans the median AR6 scenario-decade error and the range of the paper's Fig. 5a). These curves are descriptive and enter no decision.

Inference. Bootstrap with $R = 1000$ resamples, with 95% percentile intervals on all reported indices. The resampling unit is the training (outer point), each resampled training carrying its eight inner points, so that evaluations sharing a trained generator are not treated as independent.

Comparison across cases. Indices are compared with the other cases of the series, among them the analysis of Li P.-H. et al. (2023) with Jansen's estimators, in relative terms only: the ranking of the factors and each index minus the mean index of its matched dummies. Ratios to a dummy are not used, because under Jansen's estimators the dummy's index is close to zero, and under binned estimators it shrinks like $1/N$, so that a ratio grows with the sample size. For a like-for-like check, the estimators of this section, with the same dummy construction, are also applied to the base matrices A and B of the companion pick-freeze designs, which are samples of the input distribution in their own right. Absolute values remain incomparable to the extent that training noise dilutes the first-order indices and enters the totals of this case. For the 2023 case, the runs are those of the results file archived as artefact C1 (SHA-256 [TBD: SHA-256 of the 2023 results file]), whose README gives the design, the column dictionary, the rows that form the base matrices A and B, and the mapping of its twelve factors to the coordinates of G1–G8. Only those rows enter, A and B pooled (8,192 of its 57,344 runs, $N = 4{,}096$ each), with its thirteen claim indicators as outputs. Factors are matched at the level of coordinates. Only factors that the README maps to a coordinate are compared, and where it maps several to one coordinate, all are reported and the largest of their dummy-corrected indices represents them. The cells are those of this section, $M = 8$ equal-count bins for continuous factors, with $M \in \{5, 12\}$ also reported, and levels for discrete ones. Since the 2023 design has already been run, its three dummy columns are pseudo-random rather than columns of the design: `numpy.random.default_rng(2023).random((8192, 3))`, rows in the order of A and then B as the README lists them, each cut like the factor it serves. The bootstrap resamples runs. In both cases the shared coordinates are ranked by their dummy-corrected first-order indices, per claim and averaged over the claims that have indices in

that case, and the original Jansen indices at $N = 4{,}096$ are reported alongside. The comparison is computed with the deposited analysis code and enters no decision.

8 Decision criteria

The primary index for the ensemble output is the projected OT index of §7 on the marginal-quantile embedding of the three categories jointly. Its variants per category, on the sliced embedding and with the Sinkhorn divergence are robustness checks. A coordinate is influential for the ensemble output if the lower bound of the 95% interval of its primary index exceeds the upper bound of the 95% interval of the mean index of its matched dummies, which measures what training stochasticity and finite-sample estimation produce under the same estimator and cells. Ensemble indices exist for the nine expensive coordinates only, since ensembles are compared at G7's paper value and G8 does not act on them. The same criterion applies to the first-order index S_j of each claim's pass indicator, for all eleven coordinates. A factor with two coordinates (G3, G4, G5) is influential if either is. These are the primary decisions. Two secondary criteria, reported separately, read the residual. Interactions are material if the lower bound of the 95% interval of $1 - \sum_j (S_j - D_j) - T_D$ exceeds zero, where D_j is the mean first-order index of coordinate j's matched dummies. A coordinate that is not influential acts through interactions if the lower bound of its T_j exceeds the upper bound of its reference total, T_D, or T_{D4} for G7 and G8. Both readings are expected to be conservative, because the reference totals carry the largest nearest-neighbour bias, and the mock tests of §10 check this. For the ensemble outputs they use the aggregated variance-based indices of §7. δ_0 is reported as the distance below which two ensembles cannot be told apart from retrained replicates. A claim is robust if it holds in at least 80% of evaluations across the design, and fragile if it holds in fewer than 50%, judged on the point estimate, with its bootstrap interval reported. A claim that holds in all or in none of the evaluations has no indices. The study's headline question, whether the paper's findings are properties of the framework or of the analyst's choices, is answered by the joint pattern of (i) claim fragility and (ii) the identity of the dominant factors (declared versus undeclared choices, as classified in Table 1, the dominant factors being those influential by the primary criterion).

9 Deviations policy

Any departure from this protocol is logged in `DEVIATIONS.md` with date, reason, and the analysis state at the time of the change. Results affected by deviations are reported separately from the pre-registered analysis. Adjudications of placeholders completed before the Zenodo freeze are part of this protocol. The deposited PDF contains no placeholders. The authors of the target paper will be informed of the deposit and invited to comment. Any change their comments prompt is a deviation, recorded before the first run if the comments arrive before it.

10 Archive contents at deposit

Protocol PDF, `DATA.md`, `DISCREPANCIES.md`, `DEVIATIONS.md` (empty), `config/paper_values.yaml` (frozen), Phase-0 reports (`results/`, including the reports of Phases 0B to 0E), the scripts that produced them (`phase0/`) and their identifier lists, including the X5 folds (`data/derived/`), the raw-data manifest, v1.0.0/v1.0.1 diff, and the 2023 results file with its script and README (`data/external/gsa2023/`, artefact C1). GSA results: none, no claim rule evaluated at any factor setting (the paper's own values included), no retraining performed, no noise-floor replicate run. The analysis code is deposited afterwards, as a new version of this Zenodo record, before the first run, a run being any training of a generator or any evaluation of a claim rule on generated data, pilots included. That version holds the analysis harness with the commit hash it was taken from, its tests on mock data (arrays of the right shapes, generated without the pipeline),

which also measure the null values of the indices and check that the secondary criteria of §8 are conservative, and the environment lock. No run precedes it, and the Zenodo timestamps of the two versions fix the order.

A Model groups of claim X5

SI S2.3 merges model versions into frameworks, citing Dekker et al. (2023), *Nature* 624:309–316 (reference 13 of the SI), and keeps the first nine rows of SI Table S3 as the model feature of its variable ranking. The main text cites the same authors' study of model fingerprints (Dekker et al. 2023, *Nat. Energy* 8:1395–1404) for the idiosyncratic model structures that X5 tests. Table 2 gives the mapping with the AR6 v1.1 model names, as fixed in `config/paper_values.yaml`, and the scenario counts of Phase 0E. SI Table S3 spells four names slightly differently (POLES CD_LINKS, REMIND-Building 2.0, and MESSAGEix-GLOBIOM 1.1 and 1.2 without the underscore). The 901-set contains no IMAGE or WITCH scenario (D15), so X5 uses the eight groups that occur there: the seven remaining groups of the nine and GEM-E3 (row 12 of Table S3), which forms a group of its own so that every real scenario has a label. C-ROADS, EPPA and MERGE-ETL (rows 10, 11 and 13) have no scenario in the 901-set. The smallest group, GEM-E3, has 41 scenarios, enough for five stratified folds. In C78, however, AIM and COFFEE have one scenario each, so their rows of the C78 confusion matrix rest on a single prediction. This is why the uniform reference computed from the whole 901-set is reported as a variant.

Table 2: Model groups of claim X5: AR6 v1.1 model names and scenarios per population (C1234/C56/C78). The paper's nine groups in the order of SI Table S3, then GEM-E3 and the models left ungrouped.

Group	AR6 model names	1,202	1,160	901
IMAGE	IMAGE 3.0, IMAGE 3.0.1, IMAGE 3.0.2, IMAGE 3.2	68/38/36	68/38/36	0/0/0
TIAM-ECN	TIAM-ECN 1.1	26/14/5	26/14/5	26/14/5
GCAM	GCAM 4.2, GCAM 5.2, GCAM 5.3, GCAM-PR 5.3	34/7/7	33/7/4	33/7/4
POLES	POLES ADVANCE, POLES CD-LINKS, POLES EMF30, POLES EMF33, POLES ENGAGE, POLES GECO2019	64/31/19	64/31/19	64/31/19
AIM	AIM/CGE 2.0, AIM/CGE 2.1, AIM/CGE 2.2, AIM/Hub-Global 2.0	32/17/6	32/17/1	32/17/1
MESSAGE	MESSAGE-GLOBIOM 1.0, MESSAGEix-GLOBIOM 1.0, MESSAGEix-GLOBIOM_1.1, MESSAGEix-GLOBIOM_1.2, MESSAGEix-GLOBIOM_GEI 1.0	161/77/28	151/77/19	150/77/19
REMIND	REMIND 1.6, REMIND 1.7, REMIND 2.1, REMIND-Buildings 2.0, REMIND-MAgPIE 1.5, REMIND-MAgPIE 1.7-3.0, REMIND-MAgPIE 2.0-4.1, REMIND-MAgPIE 2.1-4.2, REMIND-MAgPIE 2.1-4.3, REMIND-Transport 2.1	185/53/59	185/53/58	185/53/58
WITCH	WITCH 4.6, WITCH 5.0, WITCH-GLOBIOM 3.1, WITCH-GLOBIOM 4.2, WITCH-GLOBIOM 4.4	56/33/26	56/33/26	0/0/0
COFFEE	COFFEE 1.1	34/30/1	34/30/1	34/30/1
GEM-E3	GEM-E3_V2021 (outside the nine)	30/8/3	30/8/3	30/8/3
not grouped	C-ROADS-5.005, EPPA 6, MERGE-ETL 6.0	10/1/3	1/0/0	0/0/0
Total		700/309/193	680/308/172	554/237/110